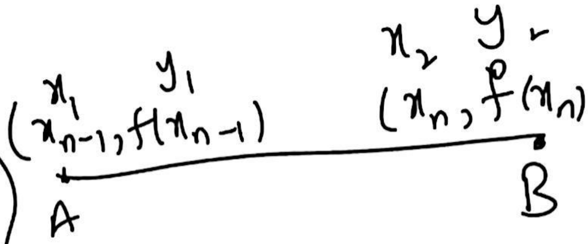
Q. Derive a formula of Secant Method.

<u>Sol.</u> We know that N.R.F is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



Slope of AB

$$= \frac{y_2 - y_1}{x_2 - x_1}$$

$$f'(x_n) \cong \frac{y_2 - y_1}{x_2 - x_1}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{\frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}}$$

$$x_{n+1} = \frac{x_n}{1} - \frac{f(x_n) \cdot (x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$x_{n+1} = \frac{x_n f(x_n) - x_n f(x_{n-1}) - f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$= \frac{\cancel{x_n f(x_n)} - x_n f(x_{n-1}) - \cancel{x_n f(x_n)} + x_{n-1}\cdot f(x_n)}{f(x_n) - f(x_{n-1})}$$

$$\boxed{x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}}$$

Q. Calculate rate of Convergence of Secant Method.

Sol. We Know That Secant formula is

$$x_{n+1} = x_n - \frac{f(x_n)\cdot(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})} \rightarrow \bigstar$$

let $x_n = a + \epsilon_n$, $x_{n+1} = a + \epsilon_{n+1}$

$x_{n-1} = a + \epsilon_{n-1}$

$$\cancel{a} + \epsilon_{n+1} = \cancel{a} + \epsilon_n - \frac{f(a+\epsilon_n)\cdot(\cancel{a} + \epsilon_n - \cancel{a} - \epsilon_{n-1})}{f(a+\epsilon_n) - f(a+\epsilon_{n-1})}$$

$$f(a) = 0 + \cdots$$

$$= \epsilon_n - \frac{(\epsilon_n - \epsilon_{n-1})\left[\epsilon_n f'(a) + \frac{\epsilon_n^2 f''(a)}{2!} + \cdots\right]}{\left[\epsilon_n f'(a) + \frac{\epsilon_n^2 f''(a)}{2!} + \cdots\right] - (\epsilon_{n-1}) \cdot f'(a) - \frac{(\epsilon_n^2 - 1) \cdot f''(a)}{2!} + \cdots}$$

$$f(a) = 0$$

$$= \epsilon_n - \frac{(\epsilon_n - \epsilon_{n-1})\left[\epsilon_n f'(a) + \frac{\epsilon_n^2 f''(a)}{2!} + \cdots\right]}{\left[\epsilon_n f'(a) + \frac{\epsilon_n^2 f''(a)}{2!} + \cdots\right] - (\epsilon_{n-1}) \cdot f'(a) - \frac{(\epsilon_n^2 - 1) \cdot f''(a)}{2!} + \cdots}$$

$$= \epsilon_n - \frac{(\epsilon_n - \epsilon_{n-1})\left[\epsilon_n f'(a) + \frac{\epsilon_n^2 \cdot f''(a)}{2!}\right]}{(\epsilon_n - \epsilon_{n-1}) f'(a) + (\epsilon_n^2 - \epsilon_{n-1}^2) \frac{f''(a)}{2!}}$$

$$= \epsilon_n - \frac{\cancel{(\epsilon_n - \epsilon_{n-1})}\left[\epsilon_n f'(a) + \frac{\epsilon_n^2 f''(a)}{2!}\right]}{\cancel{(\epsilon_n - \epsilon_{n-1})}\left[f'(a) + (\epsilon_n + \epsilon_{n-1}) \frac{f''(a)}{2!}\right]}$$

$$= \epsilon_n - \frac{\cancel{(\epsilon_n - \epsilon_{n-1})}\left[\epsilon_n f'(a) + \frac{\epsilon_n^2 f''(a)}{2!}\right]}{\cancel{(\epsilon_n - \epsilon_{n-1})}\left[f'(a) + \frac{(\epsilon_n + \epsilon_{n-1}) f''(a)}{2!}\right]}$$

$$= \epsilon_n - \frac{\cancel{f'(a)}\left[\epsilon_n + \frac{\epsilon_n^2 f''(a)}{2 f'(a)}\right]}{\cancel{f'(a)}\left[1 + \frac{(\epsilon_n + \epsilon_{n-1}) f''(a)}{2 f'(a)}\right]}$$

$$= \epsilon_n - \left[\epsilon_n + \frac{\epsilon_n^2 f''(a)}{2 f'(a)}\right]\left[1 + \frac{(\epsilon_n + \epsilon_{n-1}) f''(a)}{2 f'(a)}\right]^{-1}$$

$$= \epsilon_n - \left[\epsilon_n + \frac{\epsilon_n^2 f''(a)}{2 f'(a)}\right]\left[1 - \frac{(\epsilon_n + \epsilon_{n-1}) f''(a)}{2 f'(a)}\right]$$

$$= \epsilon_n - \left[\epsilon_n + \frac{\epsilon_n^2 f''(a)}{2 f'(a)}\right]\left[1 - \frac{\epsilon_n f''(a)}{2 f'(a)} - \frac{(\epsilon_{n-1}) f''(a)}{2 f'(a)}\right]$$

$$= \epsilon_n - \Big[\epsilon_n - \cancel{\frac{\epsilon_n^2 f''(a)}{2f'(a)}} - \frac{(\epsilon_n \cdot \epsilon_{n-1}^2) f''(a)}{2f'(a)}$$

$$+ \cancel{\frac{\epsilon_n^2 f''(a)}{2f'(a)}} - \frac{\epsilon_n^3 (f'''(a))^2}{4(f'(a))^2} - \frac{\epsilon_n^2 \cdot \epsilon_{n-1}^2 (f''(a))^2}{4(f'(a))^2}$$

ignore higher power of derivative.

$$= \cancel{\epsilon_n} - \cancel{\epsilon_n} + \frac{(\epsilon_n \cdot \epsilon_{n-1}) f''(a)}{2f'(a)}$$

ignore higher power of derivative.

$$= \cancel{\epsilon_n} - \cancel{\epsilon_n} + \frac{(\epsilon_n \cdot \epsilon_{n-1}) f''(a)}{2f'(a)}$$

$$\epsilon_{n+1} = \frac{f''(a)}{2f'(a)} \cdot \epsilon_n \cdot \epsilon_{n-1}$$

$$\epsilon_{n+1} = K \cdot \epsilon_n \cdot \epsilon_{n-1}$$

$$\epsilon_{n+1} \propto \epsilon_n \cdot \epsilon_{n-1}$$

It is called ~~2nd or~~ rate of convergent of S.M

Q. Prove That Newton Raphson Method is quadratically Convergent Method OR It is 2nd order Method.

Proof. We Know That Newton Raphson formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \rightarrow \star$$

let us suppose That $x_n = a + \epsilon_n$

$$x_{n+1} = a + \epsilon_{n+1}$$

$$a + \epsilon_{n+1} = a + \epsilon_n - \frac{f(a + \epsilon_n)}{f'(a + \epsilon_n)}$$

$$\epsilon_{n+1} = \epsilon_n - \frac{\left[f(a) + \epsilon_n \cdot f(a) + \frac{\epsilon_n \cdot f(a)}{2!} \right]}{\left[f'(a) + \epsilon_n \cdot f''(a) + \frac{\epsilon_n^2 \cdot f'''(a)}{2!} \right]}$$

$$\because f(a) = 0$$

$$\epsilon_{n+1} = \epsilon_n - \left[\epsilon_n f'(a) + \frac{\epsilon_n f''(a)}{2!} \right]$$

$$\epsilon_{n+1} = \epsilon_n - \frac{\left[f(a) + \epsilon_n \cdot f(a) + \frac{\epsilon_n \cdot f(a)}{2!} \right]}{\left[f'(a) + \epsilon_n \cdot f''(a) + \frac{\epsilon_n^2 \cdot f'''(a)}{2!} \right]}$$

$$\because f(a) = 0$$

$$\epsilon_{n+1} = \epsilon_n - \frac{\left[\epsilon_n f'(a) + \frac{\epsilon_n^2 f''(a)}{2!} \right]}{\left(f'(a) + \epsilon_n f''(a) + \frac{\epsilon_n^2 f'''(a)}{2!} \right]}$$

$$= \epsilon_n - \epsilon_n f'(a) \left(\frac{\epsilon_n f'(a)}{\epsilon_n f'(a)} + \frac{\epsilon_n^2 \cdot f''(a)}{\epsilon_n f'(a) \cdot 2} \right)$$

$$= \epsilon_n - \frac{\epsilon_n \cancel{f'(a)} \left(\frac{\epsilon_n \cancel{f'(a)}}{\cancel{\epsilon_n f'(a)}} + \frac{\overset{\epsilon_n}{\cancel{\epsilon_n^2}} \cdot f''(a)}{\cancel{\epsilon_n} f'(a) \cdot 2} \right)}{\cancel{f'(a)} \cdot \left(\frac{\cancel{f'(a)}}{\cancel{f'(a)}} + \frac{\epsilon_n f''(a)}{f'(a)} + \frac{\epsilon_n^2 f'''(a)}{f'(a) \cdot 2} \right]}$$

ignoring higher derivative

$$= \epsilon_n - \frac{\epsilon_n \left[1 + \frac{\epsilon_n f''(a)}{2f'(a)}\right]}{\left[1 + \frac{\epsilon_n f''(a)}{f'(a)}\right]^1}$$

$$= \epsilon_n - \epsilon_n \left[1 + \frac{\epsilon_n f''(a)}{2f'(a)}\right]\left[1 + \frac{\epsilon_n f''(a)}{f'(a)}\right]^{-1}$$

$$= \epsilon_n - \epsilon_n \left[1 + \frac{\epsilon_n f''(a)}{2f'(a)}\right]\left[1 - \frac{\epsilon_n f''(a)}{f'(a)}\right]$$

$$= \epsilon_n - \epsilon_n \left[1 - \frac{\epsilon_n f''(a)}{f'(a)} + \frac{\epsilon_n f''(a)}{2f'(a)} - \frac{\epsilon_n^2 f''^2(a)}{2f'^2(a)}\right]$$

$$= \epsilon_n - \epsilon_n \left[1 - \frac{\epsilon_n f''(a)}{f'(a)} + \frac{\epsilon_n f''(a)}{2 f'(a)} - \frac{\epsilon_n^2 f''^2(a)}{2 f'^2(a)} \right]$$

$$= \epsilon_n - \epsilon_n \left[1 - \frac{2 \epsilon_n f''(a) + \epsilon_n f''(a)}{2 f'(a)} \right]$$

$$= \epsilon_n - \epsilon_n \left[1 + \frac{-2 \epsilon_n f''(a) + \epsilon_n f''(a)}{2 f'(a)} \right]$$

$$= \epsilon_n - \epsilon_n \left[1 - \frac{\epsilon_n f''(a)}{2 f'(a)} \right]$$

$$= \cancel{\epsilon_n} - \cancel{\epsilon_n} + \frac{\epsilon_n^2 f''(a)}{2 f'(a)}$$

$$\epsilon_{n+1} = \cancel{\epsilon_n} - \cancel{\epsilon_n} + \frac{\epsilon_n^2 f''(a)}{2 f'(a)}$$

$$\epsilon_{n+1} = \epsilon_n^2 \cdot \frac{f''(a)}{2 f'(a)} \qquad \therefore K = \frac{f''(a)}{2 f'(a)}$$